# Eric Yang

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Machine learning researcher and engineer specializing in AI evaluation, foundation models, and multimodal learning. M.S. in Health Data Science from UCSF with experience leading research across 156,724 medical images, LLM-to-physician benchmarking, and production AI systems spanning 100,000+ multimodal music tracks. Combines rigorous experimental design and failure analysis with end-to-end Python/SQL/PyTorch implementation.

## EDUCATION

**University of California, San Francisco (UCSF)** — San Francisco, CA
M.S. in Health Data Science — Sept. 2026

**University of California, Los Angeles (UCLA)** — Los Angeles, CA
B.S. in Computational and Systems Biology, Concentration in Biological Data Sciences — Jun. 2024
Departmental Highest Honors

## RESEARCH EXPERIENCE

**UCSF Department of Medicine** — San Francisco, CA
*Data Science Specialist (Part-time)* — Apr. 2025 – Apr. 2026

**Empirical Evaluation of Large Language Models Against Human Expert Judgment**

- Led empirical evaluation of GPT-5 Pro, GPT-4.1, and GPT-3.5 across ~500 medical-student OSCE encounters scored by 4 physician evaluators, benchmarking LLM-generated assessments against expert clinical judgment.
- Designed a cross-model evaluation framework using Cohen's κ, structured scoring, semantic similarity, and natural language inference; GPT-5 Pro achieved $\kappa = 0.67$, compared with 0.62 for GPT-4.1 and 0.60 for GPT-3.5 under the same evaluation protocol.
- Identified clinical organization and presentation structure as a major source of model-human disagreement: LLMs tended to over-credit generally coherent responses that physicians penalized for failing to follow clinically actionable presentation conventions.
- Built reproducible Python evaluation pipelines and translated failure analyses into rubric refinements for an AI-assisted medical simulation and feedback platform that provided students with rapid, structured performance feedback.

**UCSF Hswen Lab** — San Francisco, CA
*Graduate Student Researcher* — Oct. 2024 – Jun. 2026

**Foundation Model Evaluation for Alzheimer's Disease Prediction**

- Led first-author research evaluating retinal foundation models across 156,724 fundus images from 78,349 UK Biobank participants, including 535 AD cases.
- Designed a controlled benchmarking framework comparing RETFound, ViT-B16, ResNet-50, EfficientNet-B3, demographic baselines, multimodal fusion, and OCT-derived clinical models under consistent participant-level splits.
- Fine-tuned RETFound (ViT-Large) in PyTorch and developed end-to-end pipelines for image QC, preprocessing, class-imbalance mitigation, participant-level training, inference, and evaluation.
- Designed anatomical ablations across full-fundus, central-retinal, and vessel-enhanced representations; RETFound achieved AUROC 0.671 on full images versus 0.658 and 0.608, respectively.
- Demonstrated that fundus features added little incremental discrimination beyond demographics (AUROC 0.710 → 0.714), while OCT-derived measures increased AUROC to 0.795, consistent with stronger discriminative signal in OCT under the evaluated benchmark.
- Presented first-author findings at AAIC 2026; manuscript under review at Nature Medicine.

**Systematic Analysis of Nicotine Pouch Discussions on Reddit**

- Analyzed 37,797 Reddit posts using LDA topic modeling and VADER sentiment analysis, identifying distinct narratives around personal nicotine use, product endorsement, and availability; found predominantly positive sentiment (53.6% positive) alongside competing narratives of harm reduction, addiction, withdrawal, and adverse health effects, supporting a Social Influence manuscript under revision following peer review.

## INDUSTRY EXPERIENCE

**BOLERO** Tokyo, Japan
*Lead Data Scientist (Remote)* Apr. 2025 – Mar. 2026

- Led a 6-person AI/data engineering team with end-to-end technical ownership of an internal music analytics platform, delivering the initial production demo in 3 weeks.
- Built multimodal ML pipelines across 100,000+ multilingual tracks, integrating librosa-derived audio features with metadata, behavioral, time-series, social-platform, streaming, and popularity signals.
- Developed and fine-tuned PyTorch multimodal neural networks for track similarity and genre-specific streaming, engagement, and longitudinal popularity modeling; used 10-fold cross-validation on the 70% training partition for model selection and evaluated final models on a held-out 15% test set.
- Built representation-learning and cosine-similarity retrieval pipelines against a reference database refreshed weekly with updated song-performance data, enabling new submissions to be benchmarked against dynamically evolving historical tracks.
- Developed a Llama 4 conversational assistant and web application used by 12+ employees and four music-group clients to upload tracks, retrieve comparable songs, inspect forecasts, and query results in natural language, reducing analytical reporting from approximately 2 days to minutes.
- Containerized and deployed the system with Docker and Microsoft Azure, integrating ML pipelines, databases, APIs, and backend services; owned full-stack implementation and production troubleshooting for a platform that remains in active use.
- Supported investor diligence and presentations with live platform demonstrations ahead of the company's ¥20M seed round.

**PITTAN** Kobe, Japan
*Chief Technology Officer* Jun. 2023 – Sep. 2023

- Led development of a non-invasive skin-analysis platform translating sweat amino-acid biomarker research into an end-to-end diagnostic and personalized skincare product.
- Built Python/Raspberry Pi 4 pipelines for remote liquid-chromatography control, data logging, peak processing, amino-acid quantification, and skin-condition scoring, collecting 500 measurements in 21 days.
- Developed neural-network models integrating sweat amino-acid and sensor-derived features from 500+ clinical samples for skin-condition assessment, achieving AUROC 0.85.

## ADDITIONAL RESEARCH EXPERIENCE

**UCLA & PITTAN** Los Angeles, CA
*Undergraduate Honors Thesis* Jun. 2023 – Jun. 2024
**Sweat Amino Acid Analysis for Non-Invasive Skin Assessment**

- Independently designed and executed an experimental study evaluating sweat amino-acid biomarkers as a non-invasive diagnostic approach using 500+ clinical samples collected with Kobe University.
- Developed and evaluated random forest and multimodal neural-network models, assessing predictive performance, clinical feasibility, and translational potential.

**UCLA Suthana Lab** Santa Monica, CA
*Research Assistant* Mar. 2023 – Jun. 2024

- Processed 1,000+ EEG recordings and developed automated MATLAB signal-processing and machine-learning pipelines to improve analytical efficiency, reproducibility, and data quality.

## MANUSCRIPTS

**Yang, E.**, Buto, P., Vashisht, R., et al. (2026). *Limits of Fundus-Based Deep Learning for Alzheimer's Disease Classification: Anatomical Ablation and Clinical Benchmarking.* Under revision following peer review at *npj Digital Medicine*.

Hswen, Y., **Yang, E.**, Han, E., et al. (2026). *Themes of Online Discussions of Nicotine Pouch Use.* Under revision following peer review at *Social Influence*.

## PRESENTATIONS

**Yang, E.**, Buto, P., Vashisht, R., et al. (2026). "Limits of Fundus-Based Deep Learning for Alzheimer's Disease Prediction: Anatomical Ablation and Clinical Benchmarking in UK Biobank." *AAIC 2026*, London, UK.

**Yang, E.**, & Wong, F. (2026). “Simulating the Expert Evaluator: Immediate AI-Driven Feedback for Medical Student Oral Presentations and Written Documentation.” *UCSF AI & Education Symposium*, San Francisco, CA.

## HONORS AND AWARDS

AAIC Conference Fellowship 2026
UCSF AGSD Summer Travel Award 2026

## SKILLS

- **Programming:** Python, SQL, R, MATLAB, C/C++, Java, JavaScript
- **AI/ML:** PyTorch, scikit-learn, LLMs, Llama 4, Vision Transformers, RETFound, CNNs, Multimodal Neural Networks, Representation Learning, Fine-Tuning
- **Evaluation:** AUROC/AUPRC, Calibration, Bootstrapping, Ablation Studies, Model Benchmarking, Cohen's κ, Semantic Similarity, Natural Language Inference
- **ML Engineering:** Model Training & Inference, librosa, Audio Signal Processing, Cosine Similarity, Data Pipelines, APIs, Docker, AWS, TensorBoard, Git/GitHub